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Sap flow low-cost sensors integrated in machine learning and collaborative IoT ecosystems for the smart irrigation of olive: testing and calibration with gravimetric data

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INTRODUCTION

- Water scarcity is a critical challenge for sustainable agriculture
- Need for efficient irrigation strategies to maintain productivity
- **Smart-Orchard Ferti-Irrigation System (SOFIS)** uses IoT and AI for accurate irrigation management



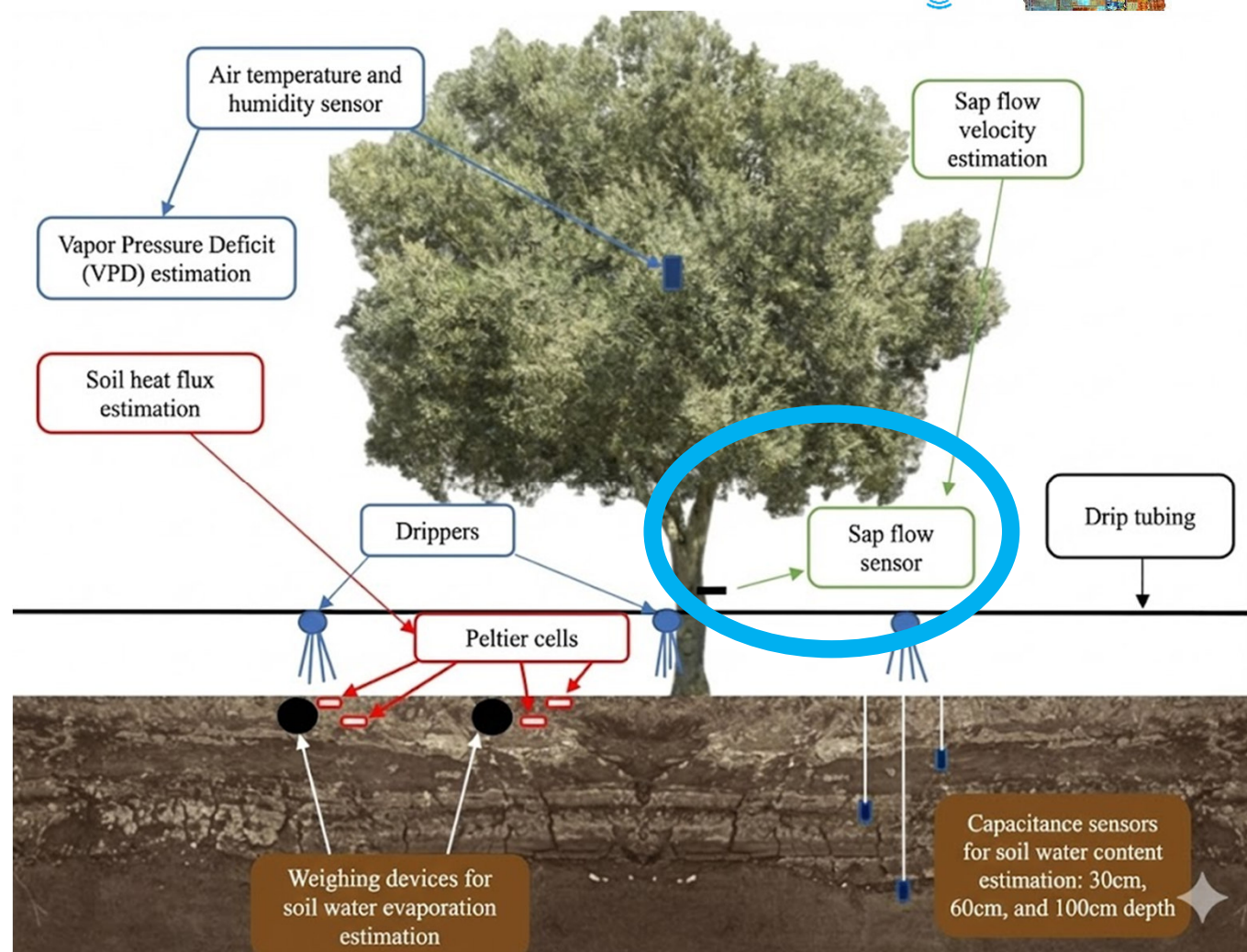
FRAMEWORK

- **SOFIS** in olive orchards – irrigation scheduling
- Compare **custom sensors** (Sap Flow, mLy, Pelt) with commercial sensors
- Assess the feasibility of **low-cost IoT technologies for farmers**

OBJECTIVES

of the present study:

- increase the accuracy of plant **transpiration** measurement
- validate the ability of the SFS to provide reliable estimates of olive transpiration
- perform SFS **calibration and testing**



Project AppiCC

Adapting Precision Irrigation to Climate Change using low-cost sensors and information technology

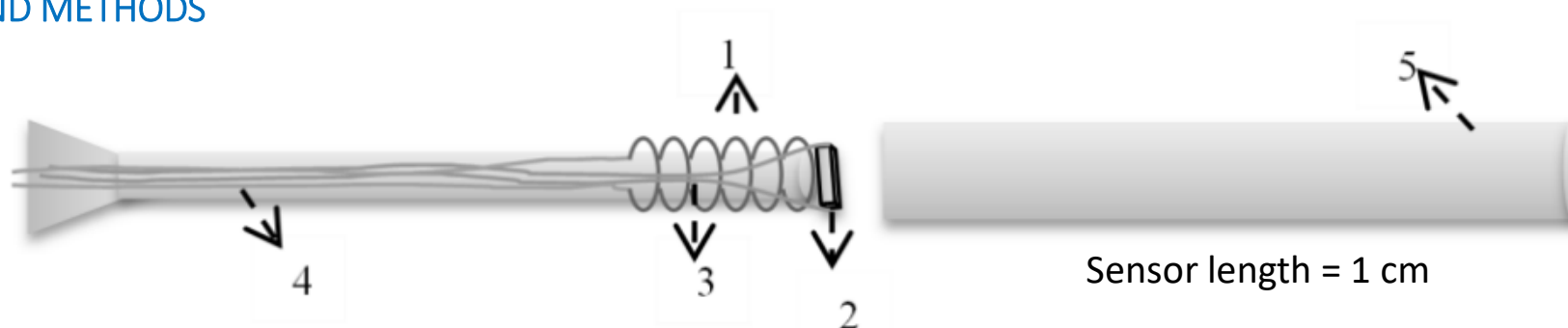
The aim

- develop a **low-cost**, customizable, modular **digital tool based on proximal sensors**
- useful in the context of irrigation management, allowing a **quick reaction** and adaptation to environmental conditions, with improved accuracy
- allow an easy transition to the **digital world** due to its low cost, open access, and specialized customer support



MATERIALS AND METHODS

Single-probe sap flow heat-pulse sensor



The modified Granier sensor design

Building components:

1. Constantan wire
2. Thermistor
3. Thermistor terminals
4. Hypodermic needle
5. Stainless tube

Calibration in controlled conditions:

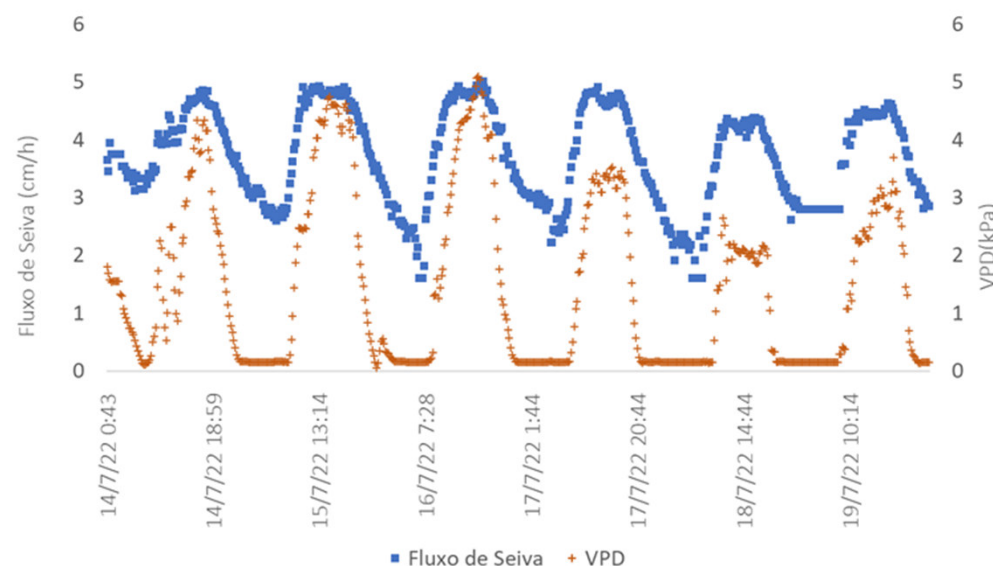
- 3 potted plants, 1 sensor each
- 3 precision weighing lysimeters
- CR3000 CS datalogger



RESULTS AND DISCUSSION

Estimation of Sap Flow Velocity

- ranged app. between 3 and 5 cm h⁻¹
- all SFS sensors exhibited comparable performance during the measurement period
- relatively low SF rates (0.02–0.03 kg h⁻¹) due to the small trunk cross-sectional area (≈ 7 cm²)
- normalized for the larger basal areas in field-grown olive trees (≈ 165 – 197 cm²), the estimated V_h would correspond to absolute SF ~ 0.5 – 1 kg h⁻¹, consistent with typical literature values of 2–12 cm h⁻¹ under comparable irrigation and evaporative conditions

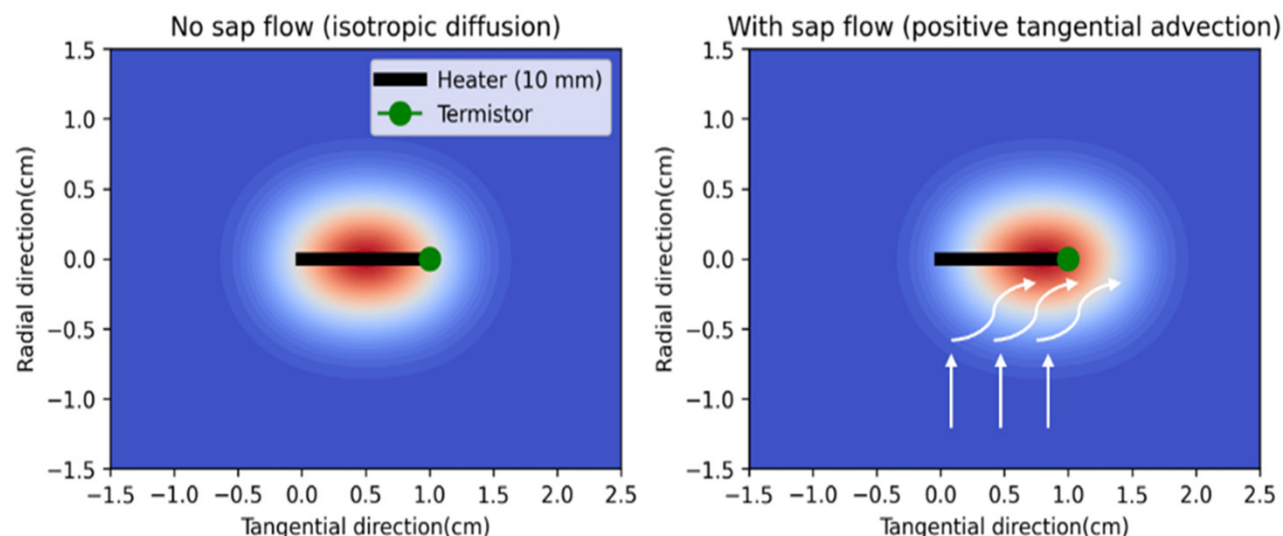


RESULTS AND DISCUSSION

Revisiting Heat-Pulse Interpretation Under Low-Flow, Small-Stem Conditions

ΔT was higher during daytime than nighttime –
opposing classical advective behavior

- Small stem under low flow, Péclet number signals **diffusion-dominated transport**
- **Sap and heat accumulation** under the sensor => when released, the thermistor senses a higher temperature than expected => ΔT
- **Daytime** $> \Delta T$ arises from lower tissue water, reduced k , and decreased pc at warmer stem temperatures
- **Nighttime** cooler, hydrated xylem promotes radial dissipation, lowering ΔT



Tangential thermal propagation in olive tree (10 mm heater, thermistor at the tip).

ΔT —the temperature rise 60 s post-heating pulse versus pre-heating baseline

RESULTS AND DISCUSSION

Sap flow sensor configuration

- **advantages** in detecting subtle local heat dynamics: it allows absolute temperature measurements at the needle tip, improves sensitivity under variable conditions, and reduces alignment errors common in multi-probe systems
- **limitations**: potential $\Delta T_{\max}$ overestimation in small stems, strong dependence on wood thermal properties (k , ρc), and sensor–tissue contact, requiring careful calibration

Relationship Between the **Thermal parameters** and Vapor Pressure Deficit

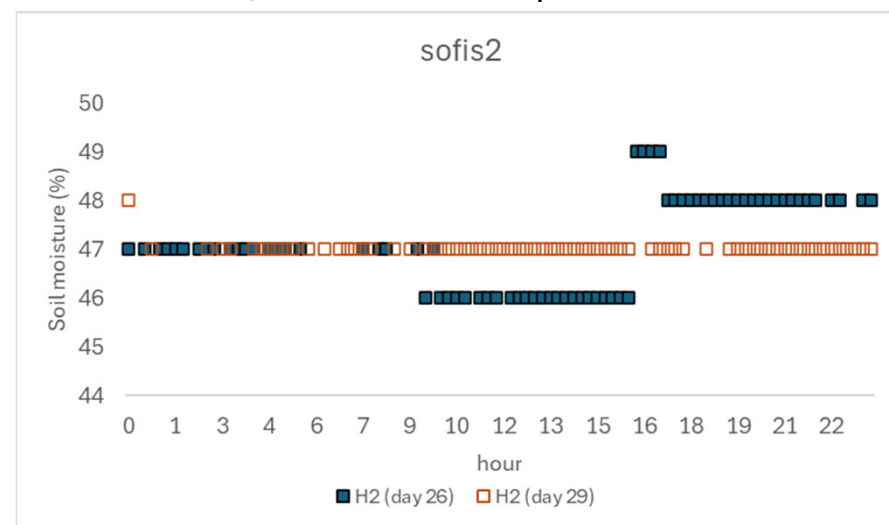
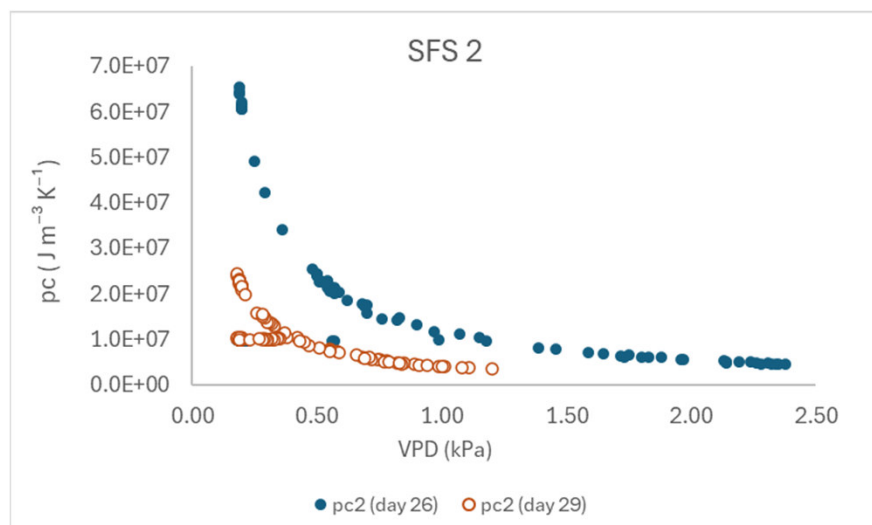
- The thermal parameters estimated from the VPD-based sap velocity model, **D** (thermal diffusivity), **ρc** (volumetric heat capacity) and **k** (thermal conductivity) are consistent with those reported in the sap flow literature for **hydrated sapwood**
- the **heat-pulse method, combined with the VPD-based velocity estimate**, is yielding realistic thermal-transport parameters for the sapwood

RESULTS AND DISCUSSION

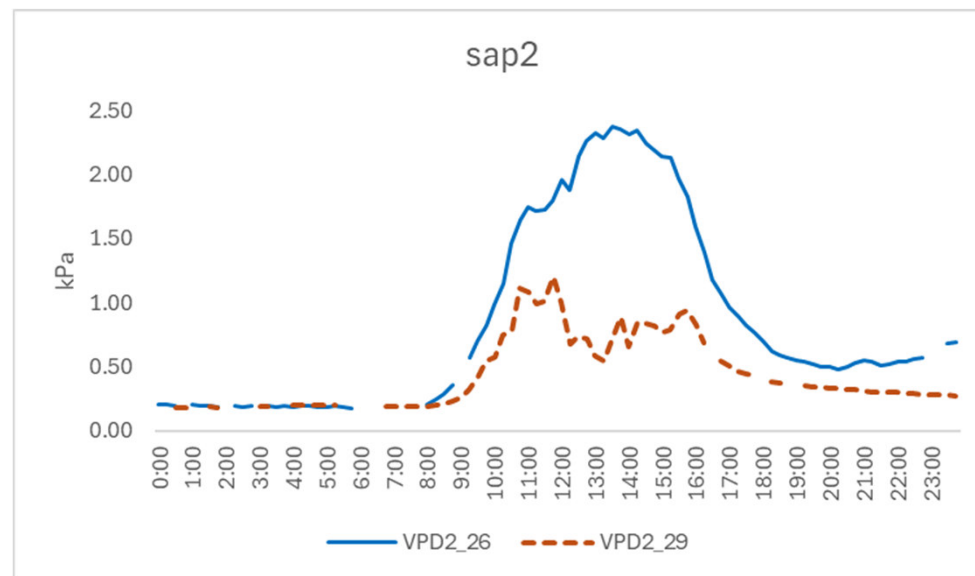
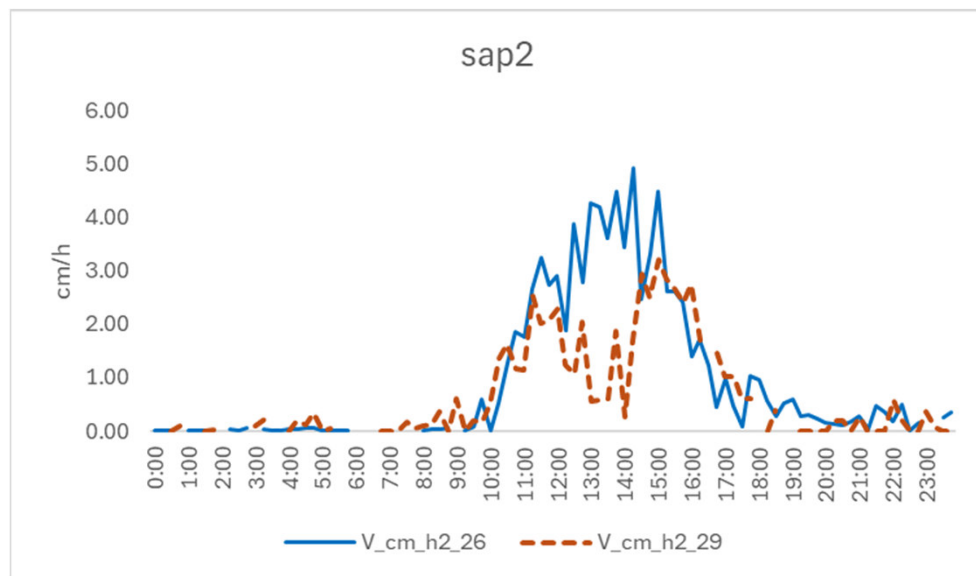
Relationship Between the Thermal parameters and Vapor Pressure Deficit

higher evaporative demand is connected to
diminished sapwood thermal storage

soil moisture held steady at 46–48% across days,
indicating well-watered plants; sap flow
variances thus stem mainly from atmospheric
drivers, not soil availability



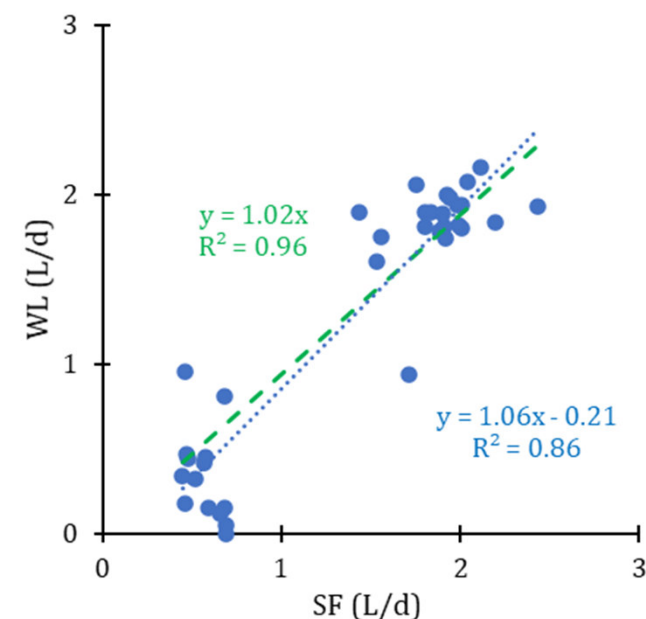
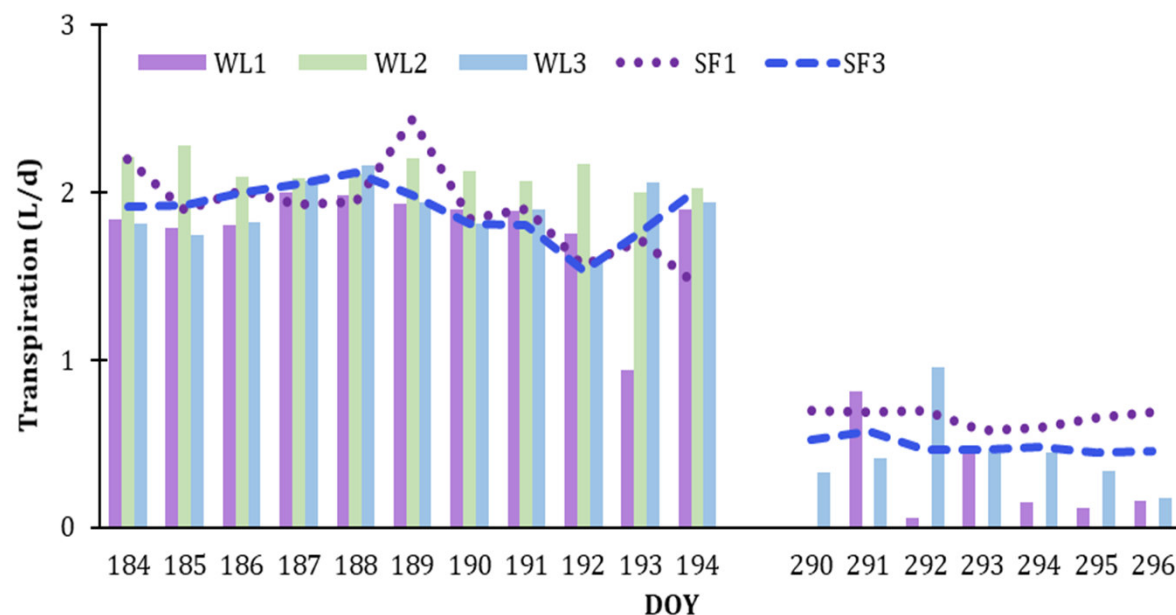
Daily patterns of vapor pressure deficit (VPD) and soil moisture, and on October 26 and 29. a) volumetric heat capacity (pc) vs. VPD; b) soil moisture



Daily patterns of sap flow velocity (V_h), and vapor pressure deficit (VPD) on October 26 and 29

RESULTS AND DISCUSSION

Calibration of sap flow sensors

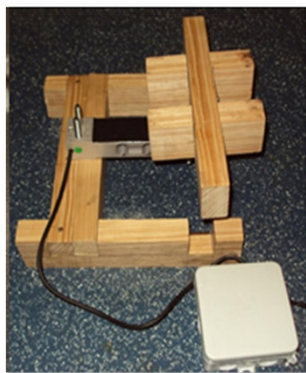


Daily transpiration measured by precision weighing lysimeters (WL) and sap flow (SF) sensors during July (184-194) and October (290-296) 2025; a) comparison; b) regression analysis; **dashed green line** - regressed to origin; **dashed blue line** - not regressed to origin

RESULTS AND DISCUSSION

Calibration of sap flow sensors

- Actual mean daily transpiration was close to 2 and 0.5 L/d per plant for the periods of July and October, respectively
- There is an underestimation by SFS below 10%, for daily values of T



Sap flow low-cost sensors for the smart irrigation of olive

Concluding remarks

- the **VPD-based advection–diffusion model** implemented in the SOFIS system could reliably estimate sap flow velocity under controlled conditions
- the system provides **real-time insights into plant water status**, allowing more efficient irrigation scheduling
- the **low cost and simple design** of the sensor makes it suitable for wider deployment, enabling continuous monitoring at multiple locations
- sap flow sensors **calibration** showed an underestimation under 10% and a high correlation to precision weighing lysimeters data
- however, **further field validation** is needed to confirm performance under variable environmental and soil conditions

Thank you!



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